Artificial Intelligence and Data Science

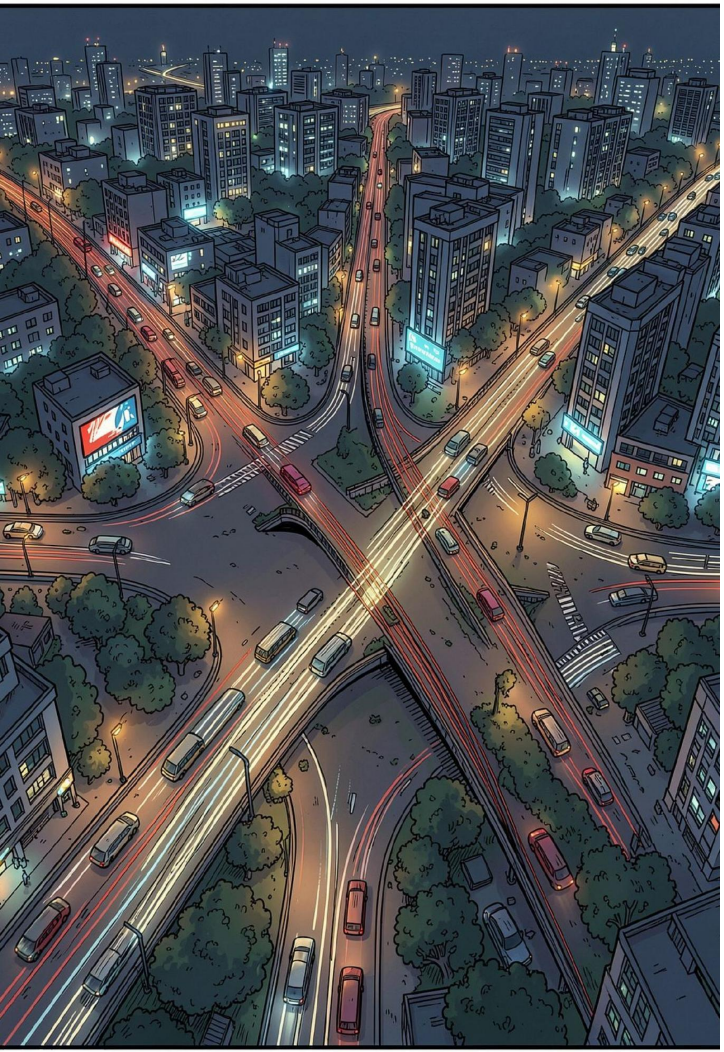
Data Science and Visualization

24ADI204

DATA-VIZ HACK-A-BIT (2026-2027)

MID REVIEW (September 2026)

Team Number: 14



India Road Accident Analysis

A Data-Driven Approach Towards Understanding Road Accident Patterns

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TEAM 14

DATA SCIENCE & VISUALIZATION · 24ADI204 · MID REVIEW

Nethra K. · Praanav Karthik M · Sri Varshini Sasikumar · Sruthi G.



The Problem We're Solving

Complex, Multi-Dimensional Data

Accidents involve time, location, vehicles, road conditions, environment, and severity — raw records alone reveal little.

Hidden Patterns Need Extraction

Systematic analysis and visualization are required to uncover trends and support road-safety planning.

Project Objectives

Understand the Data

Explore structure, types, and characteristics of **AccidentsBig.csv**.

Improve Data Quality

Handle missing values, duplicates, and outliers.

Explore Patterns

Univariate and bivariate distributions across key variables.

Analyze Relationships

Correlations between severity, casualties, and environment.

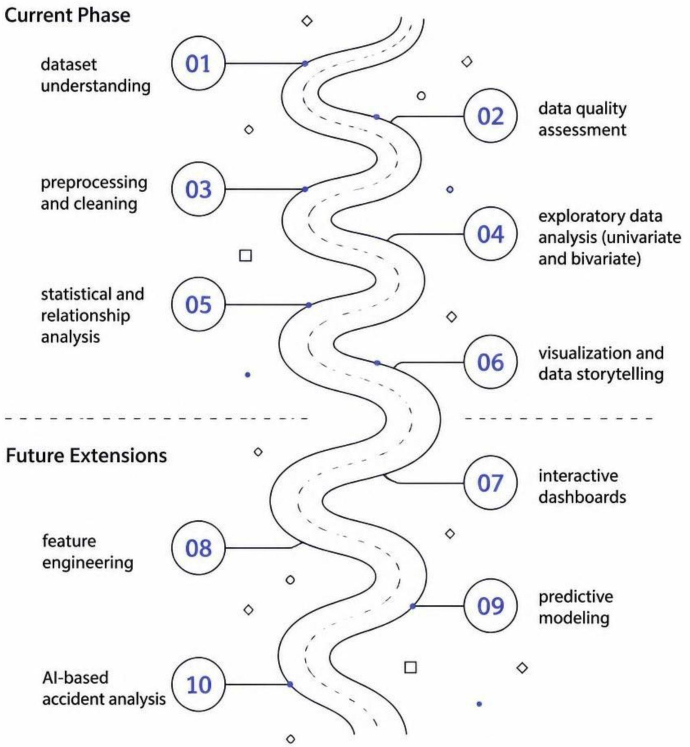
Visualize Insights

Transform findings into interpretable charts and visual narratives.

Build a Foundation

Prepare for dashboards, feature engineering, and predictive modeling.

Project Scope



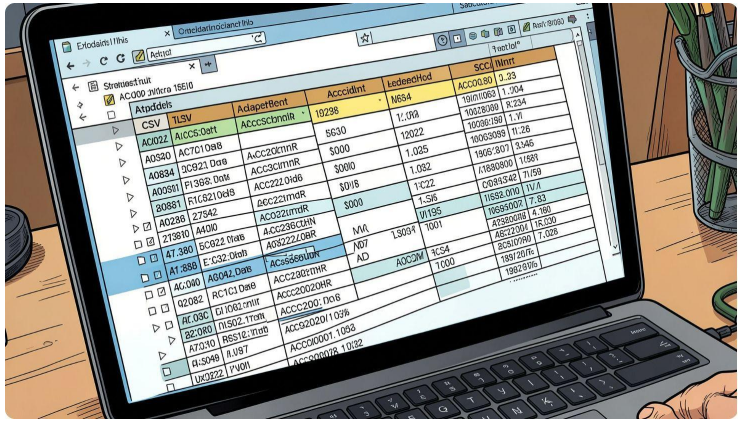
Current Phase

- Dataset understanding & quality assessment
- Preprocessing & cleaning
- Exploratory Data Analysis (univariate & bivariate)
- Statistical & relationship analysis
- Visualization & data storytelling

Future Extensions

- Interactive dashboards
- Feature engineering
- Predictive & AI-based accident analysis

Dataset Profile



AccidentsBig.csv

Structured accident dataset · Kaggle / project-provided

NUMERICAL | CATEGORICAL

Accident Characteristics

Severity, Vehicles, Casualties

Temporal

Day, Hour, Time

Road & Environment

Road Type, Speed Limit, Weather, Light, Surface

Location

Urban/Rural, Latitude, Longitude

Methodology — End-to-End Workflow

Data Collection

Load **AccidentsBig.csv**

Dataset Representation

Structure, types, dimensions

Preprocessing

Imputation, duplicates, outliers

EDA

Univariate & bivariate analysis

Statistical Analysis

Correlations & relationships

Visualization

Charts & data storytelling

Insights

Accident pattern understanding

Results — Dataset Representation

Dataset Loaded

Dimensions, structure, and data types examined

Variables Classified

Numerical and categorical columns identified

Quality Issues Found

Missing values and duplicates identified

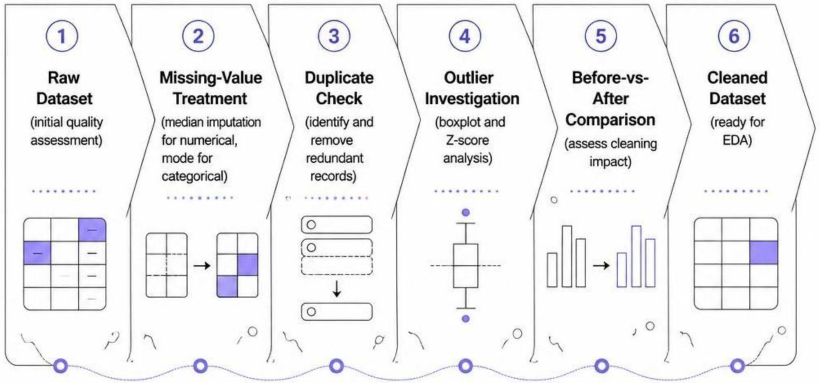
Descriptive Statistics

Summary stats and unique-value counts examined

Severity Distribution

Initial accident-severity distribution visualized

Results — Data Preprocessing



Cleaning Strategy

- **Numerical missing values** — median imputation
- **Categorical missing values** — mode imputation
- **Duplicates** — identified and removed
- **Outliers** — investigated via boxplots and Z-score
- **Impact** — before-vs-after comparisons validated

Cleaned dataset prepared for subsequent EDA stages.

Results — Exploratory Data Analysis

Univariate Analysis

- Accident severity distribution
- Vehicles, casualties, speed limit
- Day of week, hour of day
- Road type, weather, light conditions
- Road surface, urban/rural area

Bivariate Analysis

- Severity vs casualties & vehicles
- Weather, light, surface vs severity
- Speed limit vs severity
- Urban/rural vs severity
- Day & hour vs casualties
- Numerical-variable correlation heatmap

Visualization & Expected Outcomes

Visualization Toolkit

- Count plots, histograms, boxplots
- Scatter plots, grouped bar charts
- Line charts, correlation heatmaps

Visuals transform complex records into interpretable patterns.

Expected Outcomes

- Key accident patterns identified
- Temporal, road & environmental insights
- Foundation for dashboards & predictive analysis

Team Contributions



Nethra K.

Led data collection efforts and ensured data quality through robust preprocessing techniques, handling missing values and duplicates.



Praanav Karthik M.

Conducted in-depth Exploratory Data Analysis (EDA), uncovering patterns and performing statistical modeling to identify key relationships.



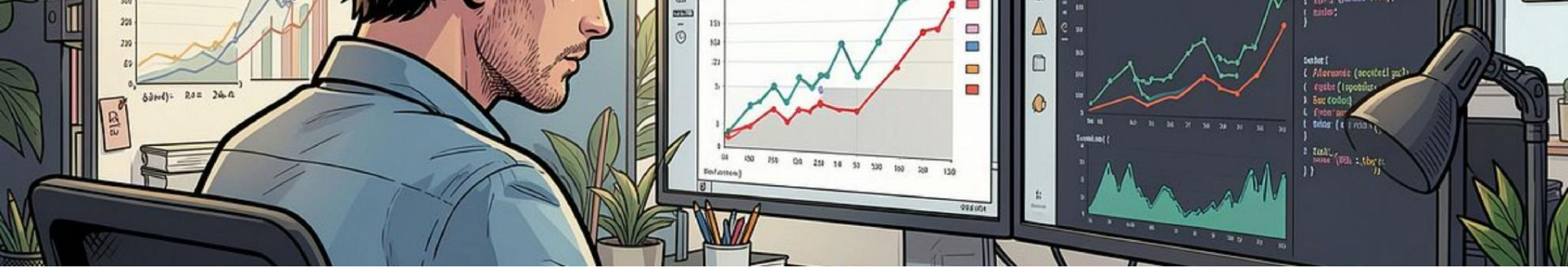
Sri Varshini Sasikumar.

Responsible for designing compelling visualizations and crafting the data storytelling narrative to communicate complex findings effectively.



Sruthi G.

Focused on extracting actionable insights and formulating strategic recommendations based on the analytical results for road safety planning.



Data Sources & Methodologies

Our comprehensive analysis is founded on robust data and cutting-edge analytical tools, ensuring reliable insights into India's road accident landscape.

- **AccidentsBig.csv Dataset:** The core dataset providing extensive road accident records for in-depth statistical analysis.
- **Kaggle Platform:** Leveraged for data sourcing, inspiration from existing analyses, and validating data integrity.
- **Python Libraries:**
  - **pandas:** Essential for efficient data cleaning, transformation, and manipulation.
  - **matplotlib & seaborn:** Utilized for generating impactful and clear data visualizations.
  - **scikit-learn:** Employed for machine learning models and advanced statistical analysis.
- **Academic & Industry Sources:** Referenced scholarly articles and sector-specific reports for contextual understanding and validation of findings.

Thank You

We appreciate your time and attention.

Our team is dedicated to uncovering actionable insights for road safety.

